Prithvi Vashisht

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Education

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science in Electrical Engineering - GPA: 3.3 — *Expected: June 2027*

Experience

Ignova Mechanical — Phoenix, AZ
Engineering Intern — *June 2025 – Sep 2025*

- Developed AutoHotKey scripts to automate end-to-end JobBOSS quote generation, parsing customer specs and populating job fields programmatically, reducing process time by 60%.
- Aided in development of a real-time PowerBI dashboard connected to live production data, enabling shop-floor visibility into job queue status, machine utilization, and scheduling bottlenecks.
- Contributed to AS9100D aerospace quality recertification by auditing and documenting engineering and manufacturing process flows against regulatory requirements.

Baskin Engineering Lab Support, UCSC — Santa Cruz, CA
Lab Technician — *April 2024 – Present*

- Diagnosed and repaired electronic hardware faults across 50+ microcontroller systems, performing component-level rework including surface-mount IC and resistor replacement using hot-air and soldering stations.
- Maintained and calibrated fabrication equipment including SLA printers and PCB mills; authored SOPs adopted lab-wide to standardize maintenance procedures and reduce equipment downtime.
- Supported 10+ active engineering courses by ensuring lab readiness, coordinating equipment availability, and assisting students with hardware setup and debugging.

Projects

Dynamic Ball Tracking in 3D Space | *MATLAB, CoppeliaSim, UR5e, Jacobian IK* — Jan 2026 – May 2026

- Designed a real-time control system tracking a moving ball using a UR5e arm in CoppeliaSim; validated on constant-velocity and unpredictable user-driven motion, achieving end-effector tracking within a 5 cm threshold.
- Implemented damped least-squares Jacobian IK with exponential position filtering and per-joint step capping; null-space projection pulled joints toward a neutral pose without disturbing end-effector accuracy.
- Delegated joint servo to CoppeliaSim's internal PD controller at 200 Hz substeps; architecture mirrors real-world systems where a high-level computer computes targets and a low-level controller closes the loop.

Autoslug Biomimetic Humanoid Hand | *Mechanical Design, Embedded C, Tactile Sensing* — Sep 2024 – May 2026

- Led a 20-member team designing a default-closed hand where servo-driven mechanisms open the fingers against a spring return force, ensuring a predictable maximum grip force and zero energy consumption while holding.
- Designed as a fully open-source module intended for attachment to the team's larger open-source humanoid platform built around the Red Rabbit Robotics torso.

Slugbotics Open-Source Modular Robotic Arm | *Mechanical Design, Embedded C, CAD* — Oct 2023 – Sep 2024

- Led a 30-member team building a fully 3D-printable modular arm; designed a preset joint-and-link system allowing the number of DOF to be extended or retracted without significant physical redesign.
- Developed control firmware in C to receive target joint angles from a host PC over UART, with independent per-joint PID loops on the onboard microcontroller driving each joint to its commanded position.
- Designed the electrical harness routing JST/servo wiring through hollow joint shafts along the axis of rotation, minimizing cable flex and wear across repeated motion cycles.
- Developed a custom quick-release end effector mount enabling tool swaps without fasteners, compatible with any end effector adhering to the defined interface geometry.

Technical Skills

Programming: Python, C, C++, Java, RISC-V, MATLAB
Software & Tools: SolidWorks, Autodesk Fusion 360, Autodesk Inventor, Git, Linux
Hardware: PCB Design (Eagle, NI Multisim), Soldering, SMD Rework, 3D Printing (FDM/SLA)
Microcontrollers & SBCs: ESP32, PIC32, ATmega328p, Teensy, Raspberry Pi
Protocols & Control: UART, I2C, SPI, PID Control
Office: Word, Powerpoint, Excel
Spoken Languages: English, French, Hindi